



Acsse CSC3B 2024 SAO1 MAIN

Computer Science (University of Johannesburg)



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FACULTY OF SCIENCE

ACADEMY OF COMPUTER SCIENCE AND SOFTWARE ENGINEERING

MODULE	COMPUTER SCIENCE 3B CSC03B3
CAMPUS	AUCKLAND PARK CAMPUS (APK)
SUMMATIVE ASSESSMENT OPPORTUNITY 1	SAO1
DATE: 2024-08-29	SESSION: 14h00 - 16h00
ASSESSOR(S):	PROF DT VAN DER HAAR MR A MAGANLAL MR R HANSLOO
MODERATOR:	INTERNAL: MR S. SITHUNGU
DURATION: 120 MINUTES	MARKS: 100

Please read the following instructions carefully:

1. You are bound by all university regulations including, but not limited to assessment, plagiarism, and ethical conduct.
2. You may not leave the venue in the first 30 minutes of the paper.
3. You may not leave the venue in the last 15 minutes of the paper.
4. Answer **all** the questions in *sequential* order.
5. Clearly indicate which question is being answered.
6. Answer only in the answer sheets provided.
7. The use of calculators is *not* permitted.
8. Write ***cleanly and legibly***.
9. This paper contains **7** questions.
10. This paper consists of **3** pages, excluding the cover page.

QUESTION 1: Operating Systems - General

- (a) **Describe** three challenges that mobile operating system designers must consider that typically do not have to be considered when designing desktop operating systems. [06]
- (b) **Name** and **briefly describe** the *hardware components* involved when you **save a file**. [08]
- (c) **Briefly describe** each of the following terms:
- i. Interrupt Vector. [02]
 - ii. Volatile memory. [02]
 - iii. Cache memory. [02]

Total: 20**QUESTION 2: Operating System Structures**

- (a) **Name** a Microsoft Windows system program that provides functionality for each of the following:
- i. File Modification. [01]
 - ii. A list of background services. [01]
 - iii. Storing and retrieving configuration information. [01]
 - iv. Scheduling tasks. [01]
- (b) **Describe** how the system call interface implements system calls on behalf of an API. [05]
- (c) **Analyse** each of the following statements. Clearly state whether the statement is **TRUE** or **FALSE** and **motivate** your answer.
- i. An operating system's *primary function* is to manage the computers **hardware and software resources**. [02]
 - ii. An operating system structure that uses loadable kernel modules is a form of a layered design. [02]
 - iii. System calls can pass parameters using a register. [02]

Total: 15**~~ Assessment continues on the next page. ~~**

QUESTION 3: Processes

- (a) **Analyse** each of the following statements. Clearly state whether the statement is **TRUE** or **FALSE** and **motivate** your answer.
- i. One program can be several processes [02]
 - ii. Processes share the stack. [02]
 - iii. All multi-processing operating systems have a short-term scheduler. [02]
- (b) **Briefly** describe what a pipe is, along with how it works within the context of processes. [03]
- (c) Answer the questions related to the following table representing process control blocks (PCBs). Each entry in the PCB takes up 4-Bytes.

State:	Waiting	Ready	Running	Ready
Name:	7x.exe	selfserv.exe	calc.exe	cmd.exe
PID:	23984	15576	17724	29020
PC:	00084256	745F28E0	007317EF	002F70F5
EAX:	0053FB94	00000000	000E6372	0005ED64
EBX:	00306000	00000000	02EB0000	03097000
ECX:	00084256	745F28E0	00731B90	002F6B20
EDX:	00084256	74400000	40140000	00000050
Addr Start:	00050000	74400000	00730000	002E0000
Addr End:	0009E000	747A1000	0073A000	0030A000

- i. **Name** the process ID that is currently being used by the processor. [01]
- ii. **Name** the process that is most likely interacting with the disk [01]
- iii. **Give** the range of memory addresses used by cmd.exe process. [01]
- iv. **Name** the address of the instruction that will be executed during the next clock cycle on the running process. [01]
- v. **Name** the process ID that may be running next, should the scheduler select The processes are according to the ascending numerical order of the process IDs. [01]
- vi. **How** many bytes are saved and loaded when a context switch happens? [01]

Total: 15

QUESTION 4: Threads

- (a) **List** three (3) benefits of using threads in an operating system. [03]
- (b) **Briefly describe** two reasons why a developer would want to create a multi-threaded web server instead of a single-threaded web server. [04]
- (c) **List** the differences between concurrency and parallelism [04]
- (d) **Critically** discuss each of the following statements. Clearly state whether you agree or not agree and motivate your answer.
 - i. A traditional process has a single thread of control. [02]
 - ii. Deferred cancellation is preferred over asynchronous cancellation. [02]

Total: 15

~~ Assessment continues on the next page. ~~

QUESTION 5: Process Synchronisation

- (a) **Describe** what a race condition is and provide one concrete example that demonstrates [05]
how it occurs.
- (b) **Discuss** the key concepts in *Peterson's Solution*, along with what problem it solves. [05]

Total: 10**QUESTION 6: Number Systems**

Show all calculation steps!

- (a) **Hexadecimal** and **octal** can be seen as shorthand for binary. **Solve** the following equation, which contains terms in multiple number systems and provides the answer in the **hexadecimal** (e.g. $q = 17_{16}$): [05]

$$x_{16} = 00001110_2 + 55_8 + 5A_{16}$$

- (b) **Convert** the number below into *two's complement* binary representation. Use exactly 8- [05]
bits to represent the number. Show all the steps in the calculation.

$$-122_{10}$$

Total: 10**QUESTION 7: 80x86 Assembly**

Provide code for 80x86 MASM assembly function that is equivalent to the following C++ function:

```
1 void flatten(int ans, int *values, int size)
2 {
3     for (int x = 0; x < size; x++) // Loop over array
4     {
5         ans -= values[x]; // subtract from ans
6     }
7 }
```

Only provide code for the function. No global variables are to be used.

Total: 15**~~ THE END ~~**